

Soln 3/9/25

Task-Note :- Utilizing 'Functions' Concepts in Python programming

Q.1 You are developing a small python script to analyze and manipulate a list of student grades for a class project. Write a Python program that satisfies the above requirements using the built-in functions `print()`, `len()`, `type()`, `max()`, `min()`, `sorted()`, `reversed()`, and `range()`.

Algorithm:

1. Start the program
2. Print welcome message: outputs a simple greeting.
3. Determine and print the number of students: uses `len()` to find the number of elements in the student - names list.
4. Print the type of lists: uses `type()` to show the type of the student - names and student - grades lists.
5. Find and print highest and lowest grades: uses `max()` and `min()` to determine the highest and lowest values in student - grades.
6. Print sorted list of grades: uses `sorted()` to sort the grades.
7. Print reversed list of grades: uses `reversed()` to reverse the sorted list and converts into a list.
8. Generate and print a range of grade indices: uses `range()` to create a list of indices from 1 to the number of students.
9. Stop.

Program:

```
def analyze_student_grades():  
    # sample data  
    student_names = ["Alice", "Bob", "Charlie", "Diana"]  
    student_grades = (85, 92, 78, 90).  
    # 1. print a welcome message  
    print("welcome to the student grades Analyzer\n")
```

output:

welcome to the student grade analyzer!

number of students: 4

type of student - names list: <class 'list'>

type of student - grades list: <class 'list'>

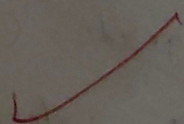
highest grade: 92

lowest grade: 78

Sorted grades: [78, 85, 90, 92]

reversed grades: [92, 90, 85, 78]

grades indices from 1 to number of students: [1, 2, 3, 4]



#2. Determine and print the number of students

```
num-students = len(student-names.)
```

```
Print("Number of Students:", num-students)
```

#3. Print the type of the student names list and the grades list

```
print("\n type of student-names list:", type(student-names))
```

```
Print("Type of Student-grades list:", type(student-grades))
```

#4. Find and print the highest and lowest grade.

```
highest-grade = max(student-grades)
```

```
lowest-grade = min(student-grades)
```

```
Print("\n Highest grade:", highest-grade)
```

```
Print("lowest grade:", lowest-grade)
```

#5. print the list of grades sorted in ascending order.

```
sorted-grades = sorted(student-grades)
```

```
Print("\n sorted grades:", sorted-grades)
```

#6. print the list of grades in reverse order.

```
reversed-grades = list(reversed(sorted-grade))
```

```
print("Reversed grades:", reversed-grades)
```

#7. Generate and print a range of grades indices from 1 to the number of students.

```
grade-indices = list(range(1-num-student+1))
```

```
Print("\n Grades indices from 1 to number of Students:",  
grade-indices)
```

Run the analysis

```
analyze-student-grades()
```

You are tasked with Creating a small calculator application to help user perform basic arithmetic operations and greet them with a personalized message. your application should perform the following tasks: addition, subtraction, multiplication, division.

Algorithm:

1. Start the program.
2. User Input for numbers: The program prompts the user to enter two numbers.
3. User Input for operation: The program prompts the user to choose an arithmetic operation. (addition, subtraction, multiplication, division)
4. Perform operation: Based on the user's choice, the program performs the chosen arithmetic operation using the defined functions.
5. Display Result: The program displays the result of the operation.
6. Stop.

Program:

```
def add(a,b):
```

```
    """
```

```
    Return the sum of two numbers"""
```

```
    return a+b
```

```
def subtract(a,b):
```

```
    """
```

```
    Return the difference between two numbers"""
```

```
    return a-b
```

```
def multiply(a,b):
```

```
    """
```

```
    Return the product of two numbers"""
```

```
    return a*b
```

```
def divide(a,b):
```

```
    """
```

```
    Return the quotient of two numbers. Handles division by zero"""
```

output:-

Arithmetic operations!

Sum of 10 & 5 = 15

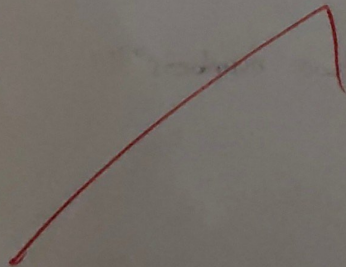
difference b/w 10 & 5 = 5

Product of 10 and 5; 50

quotient of 10 and 5; 20

greeting!

hello, alice! welcome to the program.



```

if b!=0:
    return a/b
else:
    return "Error: Division by zero"

def greet (name):
    """
    Return a greeting message for the user"""
    return f"Hello, {name}! welcome to the program".

def main():
    # Demonstrating the use of user-defined functions.
    # Arithmetic operations
    num1 = 10
    num2 = 5

    print("Arithmetic operations:")
    print(f"sum of {num1} and {num2};", add(num1, num2))
    print(f"Difference between {num1} and {num2};", student(num1, num2))
    print(f"product of {num1} and {num2};", multiply(num1, num2))
    print(f"quotient of {num1} and {num2};", divide(num1, num2))

    # Greeting the user
    user_name = "Alice"
    print("\n Greeting:")
    print(greet (user_name))

    # Run the main function
    if __name__ == "__main__":
        main()

```

VEL TECH	
EX NO.	
PERFORMANCE (5)	6
RESULT AND ANALYSIS (5)	5
IVA VOCE (5)	5
RECORD (5)	5
TOTAL (20)	
SIGN WITH DATE	15

Result: Thus the python program using 'functions' concepts was successfully executed and output was verified.